CHAPTER 8

Narrativity in Higher Order Cognition

Previous chapters have argued for a form of narrative processing to supplement research describing the role of narrative in personal identity, communication, and larger sociocultural networks. Narrative processing is ongoing meaning-directed generation and processing of emergent temporal structures consisting of related value-dynamic sequences involving value-dynamic selected agents, mediated by socio-cultural contexts. Low level narrative processing gives the phenomenology of some experiences a weakly narrative character, and it stimulates and is shaped by further cognitive processing. The previous two chapters made the case for weak narrativity in affect and cognition, locating narrativity in realms of cognitive processing far different from the autobiographical, social, and textual forms of narrative that have dominated discussions. Weak narrativity emerges via the combination of bottom-up and top-down processes in a 'reverberating causality' (see Chap. 7, Sect. 7.6).

This final chapter considers 'higher order' cognitive processes that (top-down) imbue experiences with elements of weak narrativity, and higher order qualities of experience which are shaped (bottom-up) by the weakly narrative qualities of our experiences. Higher order qualities include an ongoing sense of continuity, the deployment of knowledge garnered from prior experiences to make sense of present experiences, learning from present experiences, selfhood, autobiographical memory, and social cognition.

B. Miller, *Narrativity in Cognition*,
https://doi.org/10.1007/978-3-031-40349-1_8

This chapter thus shares with ‘small story’ research in discourse studies and work in narrative hermeneutics a view of narrative as an iterative, provisional, ongoing form of processing that continuously frames experiences within sociocultural values while imprinting sociocultural values into perception, reflection, and discourse (see Meretoja 2018; de Fina and Georgakopoulou 2011; Bamberg 2012; Brockmeier and Meretoja 2014; Georgakopoulou 2007). As Meretoja writes, “we are constituted through a continuous process of reinterpretation [that] … rarely—if ever—leads to a single life story … [but instead] emphasizes the processuality and performativity of narrative interpretation … ‘narrative’ should be understood less as a noun than as an activity” (2022, pp. 280–1).

The emphasis here, on how weak narrativity emerges in individual cognition, complements the social, interactive generation and circulation of narratives studied in discourse studies. To complete the portrait of the manifold through which seemingly individual cognitive processes generate narrativity, this chapter emphasizes higher order cognitive functions that give individuals a sense of unity and continuity through time.

Some researchers acknowledge a role for narrativity in cognition. For example, Hassabis and Maguire (2007), in the chart reproduced below, define narrative as one of nine component processes implicated in a variety of cognitive functions (Fig. 8.1).

The isolation of narrative as a discrete process (“a story structure formed by the unfolding of a sequence of events”) differs significantly from the more pervasive model of dynamic and relational valuing and temporal marking I’ve described as weak narrativity, and which is

Table I. Mapping of component processes to cognitive functions[a]

	Scene construction	Subjective time	Self	Autonoetic consciousness	Narrative	Familiarity	Visual imagery	Prospective planning	Task monitoring
Episodic memory recall	Y	Y	Y	Y	Y	Y	Y	N	Y
Episodic future thinking	Y	Y	Y	Y	Y	Y	Y	Y	Y
Navigation	Y	N	D	N	N	D	Y	Y	Y
Imagination	Y	N	N	N	D	D	Y	N	Y
Default network	Y	N	Y	D	D	D	D	N	U
Viewer replay	Y	N	N	N	Y	Y	Y	N	Y
Vivid dreaming	Y	N	D	N	Y	D	Y	N	U
Theory of mind	D	N	Y	N	N	D	D	D	D

Abbreviations: Y, yes – process is involved in that cognitive function; N, no – process is not involved in that cognitive function; D, depends – process involvement depends on the precise nature of the task and the content operated on; U, unknown – unclear if process is involved in that cognitive function.
[a]Processes are labelled along the top and cognitive functions down the left hand side.

Fig. 8.1 Mapping of component processes to cognitive functions (rpt. from Hassabis and Maguire 2007) implicated in many of the other component processes. This grouping of processes points, in particular, to problems around unity (scene construction), intentionality (prospective planning), and self (subjective time, self, familiarity, autonoetic consciousness) that provide a bridge between weak, unreflective narrativity and the highly narrative reflections of autobiographical memory and selfhood.

8.1 Unity

To explore connections between weak narrativity and more reflective cognitive functions, we'll begin with considering how weak narrativity might contribute to the phenomenology of our conscious experiences of ourselves. Part of the argument for weak narrativity is that it helps explain the narrative features we associate with experiences when we recall or report them.

These features include temporal boundaries set by patterns of equilibrium-disequilibrium-equilibrium, peak and end/fact experiences, spatial and agentive boundaries set by relational valuations, and sequential bindings driven by relations (typically causal) but heavily mediated by context (see Chaps. 6 and 7). Emergent narrativity is provisional and fragmentary, rather than closed form and formulaic.

In this model valuation is crucial and contingent upon a variety of experiences and knowledges. It is indistinguishable from perception and emotion, and it is derived in part from the subject's goals. That is, emergent narrativity seems to be part of the subject's grasping of experience as it unfolds, and it is part of the 'seeming' of experiences. This condition points to the implication of narrative processing in some key features of consciousness that might appear to have nothing to do with narrativity: phenomenality, unity, and selfhood.

Regarding phenomenal, subjective experience, the affective, embodied valuations characterized in theories reviewed in earlier chapters have parts to play as explanans in robust explanations. That is, the mysterious causes and mechanisms of "what it's like" to experience are linked to the phenomenology of weak narrativity insofar as it is grounded in emotion and prediction. The working through of these forms of processing is not a matter of abstracted calculation performed by a disembodied homunculus but is rather a visceral going through of experience with necessarily phenomenal properties. Dynamic, relational valuing is a form of knowing integral to experiencing. Because of valuation's affective, ecological, and predictive character, the 'what' of phenomenological features of experience might be considered from this perspective: How might the subjective peculiarities of an experience be related to a view of knowing that entails unfolding, dynamic, affective, relational valuing and predictive processing?

When the phenomenal aspect of consciousness is conceived in this manner, the nature of unity and selfhood appear to be driven by these processing features. The apparent unity of consciousness provides a vexing problem for scientists who recognize the distributed nature of cognitive processes. Unity, according to Tim Bayne's 2010 model, reflects the 'mereological' (i.e., part to whole) nature of any particular experience as a conscious state subsumed by the larger phenomenal state of the subject. That is, unity is a phenomenal feature of consciousness that results from the seamless switching between different processing faculties in the brain, and as such reflects only part of experience at any given moment.

On the view advanced here, the unity of consciousness is deeply related to ongoing dynamic valuation processes. Because these processes are relational and dynamic, unity is experienced as part of the synthetic operations that dominate attention. But it is intuitively possible that multiple, competing valuation processes can be ongoing simultaneously, as when one attends to present events while, 'in the back of one's mind,' as the phrase goes, one simulates and evaluates a past or anticipated event. Several theories of cognitive processes are conducive with this perspective. For example, Global Workspace Theory (see Chap. 5, Sect. 5.1) describes ongoing parallel distributed processing at a sub-personal level as well as conscious experience dominated by global broadcasting of whatever is presently most salient.

In a related view, Attention Schema Theory (see Chap. 6, Sect. 6.2) describes awareness as a partial model of our attention. In this view, the unity of consciousness reflects the salience-based apportionment of attention. We are endowed with limited capacity for attending, and the experience of phenomenal unity is similar to current descriptions of the unity of the visual field. In vision, the visual field includes space surrounding the focal area. Partial unity comes from this consistent structure. Moreover, visual unity is produced by saccades which assist in filling out the sense of a unified visual field, and the field is filled in via internal cognitive processes (rather than with additional perceptual data). This permits an experience of visual unity without an overload of perceptual data, and it allows the synthesis of data to occur in conjunction with construction.

Phenomenal unity is an especially challenging problem because it is multimodal and streaming. That is, the fact that experience feels unified seems contradictory to the different sensory inputs we receive and the changeability of these inputs through time. The 'seeming' of conscious experience is particularistic and an irreducible combination of qualities associated with vision, hearing, proprioception, internal reflection, and other factors. However, if the lesson of cognitive-emotional and predictive processing is that experience is constructed in part from valuation, then phenomenal unity is created by dynamics of attentional selection as the subject makes sense of the most salient valuation dynamic confronting the subject.

Moreover, if we think of attention as a means of amplifying or specifying valuation as well as being intrinsically related to salience (i.e., if attention is a limited resource most likely devoted to tracking salient value dynamisms for maximizing predictive efficacy), then the phenomenology of the subject's experiences is structured by the various types of valuation (see Chaps. 6 and 7) associated with the emotional, predictive, and embodied processes of cognition.

Within this model, the temporal extension produced by ongoing prediction is partly constitutive of unity. That is, a sense of unity is produced out of the modeling and parsing that accounts for and predicts change, limiting the salient agents and objects synthesized in any particular instant in the phenomenal field. Predictive modeling is not a discrete analytic process like the core sampling used by scientists to study various materials. Rather, it suffuses the unfolding of present time and *is* part of the experience.

That is, while prediction error minimization applies to object recognition, it also applies to change. Expected change confirms models of experience, while unexpected change challenges those models and stimulates additional processing to render the change coherent. So, the error minimization process of predictive modeling operates not outside of time, but while time and experience are unfolding, and therefore manifests as predictions about past, present, and future simultaneously. As the research on event segmentation demonstrates (see Chap. 6, Sect. 6.4), this dynamic coherence is not unstructured: contingent, meaningful temporal strands emerge in our experiences. In other words, the emergent temporal structures of narrative processing. The transitions between these strands, then, involve reorientation that feels part of experiential unity.

Also, phenomenal unity is likely correlated to affective tone. The constituents of a seemingly unified general affective tone differ from one another. On the one hand, emotional mood shapes the subject's affective posture toward a particular situation, and vice versa. On the other hand, the valuation of constituents of experience and their relations are both contributors to and shaped by affective tone.

These co-relations suggest a key aspect of how the meaningfulness associated with some temporal extension emerges: perceptual experiences and self-experiences are overlapping. In fact, individual valuations are synthesized into, and are shaped by, their relation to the experiencing subject.

8.2 For-Me-Ness

The models of cognition reviewed in earlier chapters—the predictive processing model (PPM), the cognitive-emotional model, and the embodied model—are characterized by dynamic valuation, which entails self-referential elements. In the case of the PPM, the experiencing is characterized by a 'making familiar or unfamiliar' of features of experience. In the case of the cognitive-emotional model, it involves hedonic feelings such as threat (at the charging attacker) or anticipated pleasure (at the arrival of an inviting plate of food at a favorite restaurant). It also involves more complex feelings derived from longer term evaluations of particulars or types manifested in experience and from socially and personally oriented schema. Finally, the embodied cognitive framework builds upon the notion of experience as an ecology of affordances—that is, what is experienced is determined by potential actions of the subject on its various affordances.

In each case, the phenomenology involves how valuation foregrounds a relation between features of experience and the subject, between some 'not-me' and 'me.' These self-oriented phenomenal properties of valuation are described in philosophy of mind in terms of 'for-me-ness.' Dan Zahavi and Uriah Kriegel, originators of this term, claim that "all conscious states' phenomenal character involves for-me-ness as an experiential constituent" (Zahavi and Kriegel 2015, p. 37). Zahavi and Kriegel do not view 'for-me-ness' as "a self-standing quale" (p. 38). This is due to its nature: "for-me-ness is not in the first instance an aspect of *what* is experienced but of *how* it is experienced; not an object of experience, but a constitutive manner of experiencing" (p. 38). Zahavi and Kriegel contrast their position to the views of Jesse Prinz and Fred Dretske, who differently point to the way experience is always *about* something and that the phenomenology of experience entails either a form of introspection (Dretske) or simply the quale presented to the subject (Prinz). Prinz believes "we always experience the world from a perspective or point of view" (p. 38), but sees this as a metaphysical fact, rather than as phenomenological (p. 39).

The view of a phenomenology of valuation advanced here seems to correspond partly to Zahavi and Kriegel's view of a 'manner' or 'how' of experiencing. Whereas they argue that "there is lemon-taste-for-me-ness, mint-smell-for-me-ness, and many other types of phenomenal character," the view advanced here sees valuation as intrinsic to, rather than a supplement of, perception (p. 38). However, this difference might involve the subject's ability to maintain some consciousness of the signatures of her own experiential historical and current context in ongoing valuation of the utility, valence, and familiarity of elements of experience.

The nexus of valuation and for-me-ness from a narrative processing perspective bears upon the experience of having a perspective. Perspective is at the heart of why some theorists have viewed narrative as essential to experience (see Chap. 2, Sect. 2.3 and Chap. 3, Sects. 3.3 and 3.4) and to our complex capacity for Theory of Mind. Yet the emergence of perspective can be located in basic cognitive processes.

8.3 Self-Specification and the Minimal Self

A number of studies shed light on how basic cognitive operations produce a minimal experience of self that forms the basis of a hierarchical model of selfhood. Christoff et al. (2011) describe reafferent-efferent circuits (including sensorimotor integration, homeostatic processes, and cognitive control, including emotion regulation) that "specify the self as a bodily agent," specifically by continually producing the distinction between self and not-self (p. 105). As opposed to common views that self-feelings are suppressed during focus-intensive tasks, they hypothesize that self-specification is integral to experience: "such tasks can be expected to enhance the self-experience of being a cognitive–affective agent" (p. 109). Indeed, agency is phenomenologically experienced within the faculties of attention and self-regulation demanded by such tasks:

> the self-experience of being an emotional agent that these processes elicit would occur at the level of affect and action tendencies [26], whereas this bodily level would be subsumed by the self-experience of being a cognitive–affective agent in deliberate emotion regulation, analogous to the way the self-experience of being a cognitive agent also subsumes the self-experience of being an embodied agent in attention-demanding cognitive tasks. (p. 110)

Affect is integral to self-specification. While Christoff et al.'s model describes how affect generates the experience of a self-agent, other research notes that emotional attitudes toward memories mark the designation of self-and not-self:

> during the recall of field (first person perspective) memories [researchers found] … a higher degree of subjective emotionality, (sic) associated with those memories in comparison to those that are retrieved from a third person ("theatrical" or "disembodied" perspective (Northoff et al. 2006). This argument echoes previous suggestions that 'the boundary between self and not-self is one's emotional attitude about an object or thought' (Barressi, 2002, qtd. in Northoff et al. 2006, p. 453). (Markowitsch and Stanilou 2011, p. 21)

These dynamics are closely related to theories that describe an inchoate self, a pre-reflective self-awareness, an illusion produced at the center of narrative gravity, a virtual selfhood, and finally a minimal selfhood.

The persistence of pre-reflective self-awareness through time constitutes a disposition toward continuity basic to the organism's interface with the world, and productive of a primal self-image. In structuring experience into meaningful episodes, the subject simultaneously generates a world oriented around a perspectival center. This creates a learning mechanism that helps the subject establish relative dispositional advantages toward types of situations.

The emergence of this self-reflective mechanism consolidates self-specification into a nascent sense of self. In this model, the emergent self is caught between pre-reflective self-awareness and self-objectification. In addition, the sense of self is subject to multiple temporalities, including an experience of the present that includes memory and anticipation, episodic and situationally based dispositions, and ultimately cumulative self-valuations that amount to an emergent autobiographical self-understanding. Each of these temporalities is subject to configuration and the reciprocal influence of other temporal "levels." As a result, self-image is both continuity-oriented and highly iterative.

Embodied cognition also points to the role of temporally unfolding experiences in fostering an emergent sense of selfhood. Evan Thompson argues for two sources of a sense of continuity: our complex time sense and a pre-reflective form of self-consciousness. Pre-reflective self-awareness can be simply described as the continuous feeling of experiencing. Thompson refers to Dan Zahavi's description of this feature of consciousness as simultaneously static and dynamic: "pre-reflective self-awareness is streaming because it is constitutive of the streaming or flowing experiences themselves, not a pure and empty awareness that appears on its own. By the same token, it is standing because it is an ever-present and unchanging feature of consciousness" (2007, p. 328). The persistence of the flow of experience, in other words, marks our presence to ourselves. This awareness provides a fundamental sense of continuity through time, even though it is not a repository of the characteristics of self. Rather, it is the perspective around which experience is organized. Thompson notes Naomi Eilan's similar model of "perspectival awareness" in terms of this implication: "such awareness is not yet 'the capacity for detached reflection on oneself' that develops along with language and conceptual thinking, but it is enough to suggest a kind of ladder or continuum between bodily interaction with the world and developed reflectivity" (p. 20).

Other philosophers and psychologists describe the link between self-specification and selfhood in terms of illusions generated during ordinary cognitive processes. In Daniel Dennett's view, we mistake a feature of processing in consciousness as a substance. The self is "a center of narrative gravity," or as Susan Schneider puts it, "a kind of program that has a persistent narrative" (Dennett 1992, p. 416; Schneider 2007). Thomas Metzinger also describes the emergence of self from a feature of consciousness processing but based upon an embodied model. In his metaphor of the "ego tunnel," Metzinger argues that conscious experience is "a low-dimensional projection of the inconceivably richer physical reality surrounding and sustaining us … Therefore, the ongoing process of conscious experience is not so much an image of reality as a tunnel through reality" (2010, p. 6). This "tunnel" reflects the transparent operation of what Metzinger terms the "phenomenal self model" (PSM)–the notion that we have a virtual model of ourselves as whole. The content of the PSM is the Ego: "the Ego is a transparent mental image: You—the physical person as a whole—look right through it. You do not see it. But you see with it" (2010, p. 8).

Timothy Bayne locates the self as a "'phenomenal fiction' generated by the self-representational field." He offers a

> quasi-realist view of the self, according to which the self is a merely virtual entity brought into being by the *de se* structure of the phenomenal field. This view, which I dub 'virtual phenomenalism', ensures an essential connection between the subject of experience and the unity of consciousness: because the subject of experience is nothing but a 'phenomenal fiction' gen-

> erated by the self-representational structure of the phenomenal field, there is an *a priori* guarantee that any experiences that a subject has at a given time will be phenomenally unified with each other. (2010, p. 2)

These frameworks position narrativity as an effect of continuity processes, rather than viewing continuity as in part generated by iterative, emergent weak narrativity. It is possible that the relationship is mutually constitutive: low level narrative processing contributes to a dynamic architecture of experiential flows that generate continuity experiences and resolve in higher order narrative processing. Minimal selfhood has been explicitly linked to our capacity to view our experiences as narrative. Like Hutto and Thompson, Richard Menary insists that the 'minimal self' is an embodied subject and agent. As described earlier (Chap. 3, Sect. 3.2), this embodied self is pre-narrative: "our embodied experiences, perceptions, and actions are all prior to the narrative sense of self, indeed our narratives are structured by the sequence of embodied experiences" (p. 75).

Menary sees these experiences "taken up into inner dialogue," which brings a new component of self into being—the discursively based narrator—which "emerges in terms of narratives as anchored in the pre-narrative sequence of experiences of an embodied subject" (p. 79). For Menary, the interposition of a discourse-based narrator is key to explaining the "intersubjective" nature of selfhood. He sees inner speech as key to generating self-awareness, and self, and discredits the idea that self is "an independent substance with intrinsic properties or a fictional object" (p. 79).

In these contrasting models of minimal selfhood, self is a fiction generated either from the ordinary operations of cognition or from its processing in inner speech. In both cases, self emerges in relation to narrative structures. If minimal selfhood is dependent upon weak narrativity or the narrative form of inner speech, and if these are influenced by and shapers of more abstracted, extended, and consciously wrought narrativity usually thought of as higher order capacities, then other recognized forms of selfhood may be grounded in narrativity.

8.4 Informational Sources of Valuations, Predictions, and Simulations

In the push-pull of integrated top-down/bottom-up processing, previously learned information shapes perception in a variety of ways. In turn, experiences produce new learning. How this information is processed provides a crucial linkage between the immediate, largely sub-personal experiences of perception and self-specification and the reflective, autonoetic experiences associated with autobiographical rumination and deliberation. How are the concepts and values that manifest in both forms of experience related and shared?

This is a profound and much debated question, and it is complicated by differences between particularistic and generalized information and questions about how they co-exist and mutually shape one another. Understanding these relations is particularly challenging for the temporal features of events, which clearly draw upon generalized, conceptual information for sensemaking, but also manifest as particular, time-stamped and phenomenologically rich structures of experience. The previous two chapters considered how events emerge via our attentiveness to emergent dynamic, relational salience networks. Here the top-down processes that shape (and are shaped by) these dynamics are considered in more detail.

A number of intermediate structures that mediate these functions have been proposed, including frames, scripts, stories, situation models, schema, and connectionist networks (see McRae et al. (2019, p. 3)). Schemas have been subject to considerable empirical investigation, so in this section we'll review this research before considering connectionist views. According to Ghosh and Gilboa,

> Schemas are general, higher level constructs that encompass representations of the similarities or commonalities across events, rather than the specificity that makes those events unique" (p. 106). Schemas are not, however, simple networks of related concepts. Instead, they "bind multiple features that consistently co-occur," are hierarchically composed of sub-schemas, include chronological relationships (Schank and Abelson), and may include action plans. (Gilboa Marlatte 2017, p. 618; Ghosh, p. 109) (p. 106)

Schemas develop and evolve via interaction with new information. In the diagram below, Ghosh and Gilboa describe how episodes create a schema and, subsequently, how a new episode leads to an adapted schema (Fig. 8.2):

Note that within Ghosh and Gilboa's figure the episodes that contribute to the schema differ in three key ways: first, the number of nodes changes (in episode l, a new node is added); second, the associative connections between nodes change (not all individual episodes feature connections between all and the same nodes, and the episodes differ in terms of the strength of connections between nodes); finally, the values of particular nodes change, indicating the necessary flexibility for associative connections to be made across nodes.

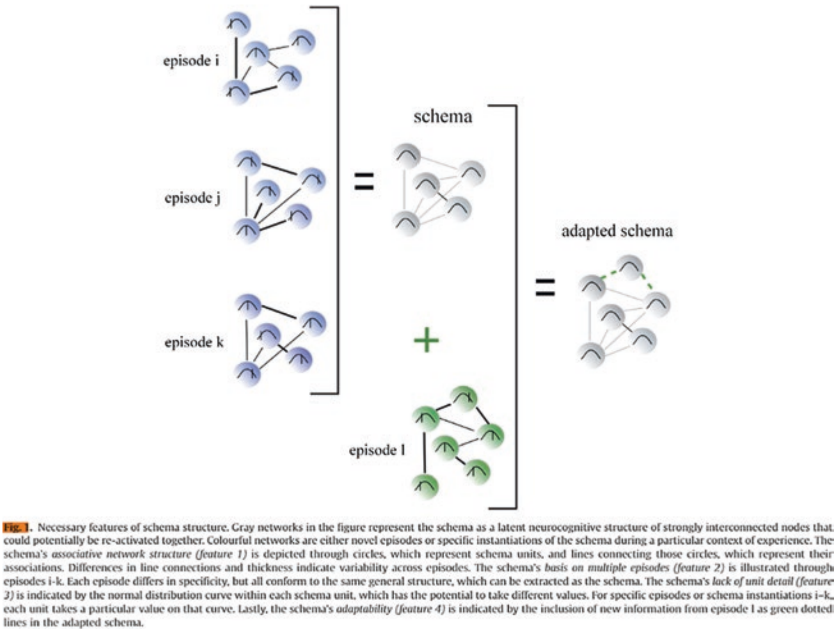

Fig. 8.2 Necessary features of scheme structure (rpt. from Ghosh and Gilboa 2014)

Schemas operate as "templates against which new information can be compared" (Ghosh and Gilboa p. 106). This may facilitate the predictive processes involved in perception. Perceptual processing with schemas likely operates through analogical recognition (what is something like, rather than exhaustive analysis of the input's properties) which activates associative networks. Bar proposes that these networks "presensitize representations of what is most likely to occur and be encountered next" (Bar 2011, p. 14). Moreover, these activated associative networks determine a "mindset," which is "a sustained (though updatable) list of needs, goals, desires, predictions, content-sensitive conventions and attitudes" (Bar 2011, p. 19). Bar provides the example of how predictive processing might engage associative networks as the subject recognizes the driver's seat of a car. The "analogies→associations→predictions" circuit activates processes that recognize other parts of a car, as well as the activity of driving.

While in this example object recognition leads to more object recognition and to action readiness, Bar notes that the model needs to account for greater temporal complexity, including "complex scenarios involving simulations and multiple elements" (p. 22). Bar's schema, in other words, "integrate[ ] representations from different points in time" in the cause of simulating an anticipated event, but it does not adequately explain how they are integrated (p. 23). In addition, schema models can struggle to explain how simulations can involve detailed particulars rather than general conceptions (as in the unweighted nodes in Ghosh and Gilboa's diagram). Because Bar's theory is linked to predictive processing, the stakes for explaining the origins of simulations is high, as it proves fundamental to some of the higher order cognitive processes considered later in this chapter.

One variant of schema theory addresses these temporal issues by positing the proliferation of 'schema instances,' or active copies of schemas into assemblages that integrate representations of an episode through time (Arbib 2020). In this model, schematic representations of experiences retain links to particular details. Arbib links schematic processing to the cognitive processes involved in navigation, which require maintenance of links to environmental particulars in structured temporal representations.

However, connectionist researchers offer persistent criticisms of schema theories: "the need to select among a pre-specified set of alternative structure types forces semantic representation into an ill-fitting procrustean bed" (McClelland, pp. 8–9). Connectionist models argue that "conceptual knowledge can emerge from a constrained learning process, without prior domain-specific knowledge and without requiring pre-specification of possible knowledge structures or selection among them" (p. 8). The connectionist approach focuses on the weightings between particulars as sources of identification and learning. In this sense, it seems more conducive to the view of valuation considered in the dimensional model of emotion (see Chap. 7).

One difference between schema accounts and connectionist accounts involves how they explain temporal structure. Whereas schema accounts include chronological relationships in the field of represented nodes, in connectionist accounts temporal structure is described as "an emergent property of a computational system that implicitly represents … time in its processing" (McRae Brown Elman, p. 215). This view is compatible with the argument for emergent temporality in event processing in Chap. 6, in which valuation dynamism structures time. However, it should be stressed that such forms of temporal emergence do not preclude their *post hoc* representation.

The connectionist critique of schema theory raises additional questions around the relation of schema (which are dynamic, but general) to several forms of fixed, but specific information structures: narratives, event gists, and episodic memories. Recent research supports the idea that "a given experience may have multiple representations, each advantageous in a different scenario" (i.e., in different forms of cognitive processing) (Schlichting et al. 2015, p. 7). Gilboa and Marlatte propose a regular process of splitting of detailed 'gist' extraction and schematization of experiences (Fig. 8.3):

Gists are abstracted elements of narrative. Ghosh and Gilboa, citing Moscovitch, define gists as 'high-level story elements that are central and critical to the overall plot'; they 'contain multiple units and their interrelationships,' and 'refer to a specific event … not based on multiple episodes' (p. 110). Ghosh and Gilboa's definition, drawn from Thorndyke, contrasts gists with unimportant story details.

Narratives, in this model, are:

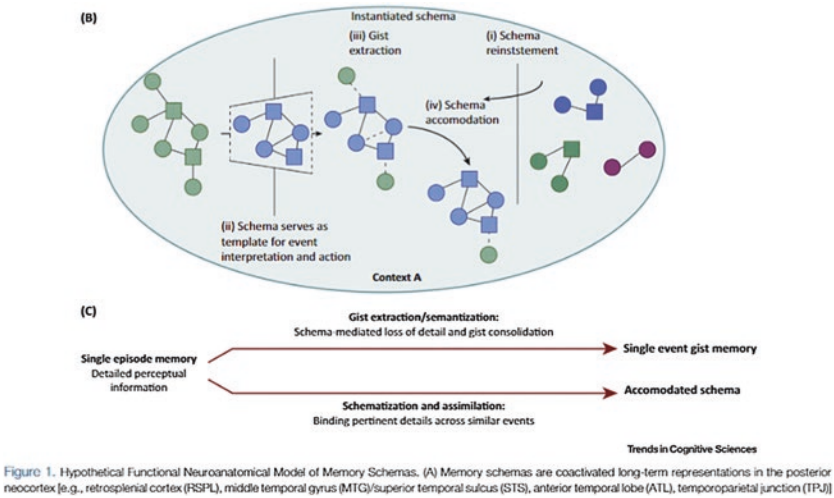

Fig. 8.3 Hypothetical Functional Neuroanatomical Model of Memory Schemas (rpt. from Gilboa Marlatte, p. 620; note: a brain diagram at the top of this figure is not included)

> series of actions and events that unfold over time, according to causal principles. As such, to be considered a narrative, both an associative network structure and chronological organization are necessary. Unlike schemas, however, in which units are lacking in detail, narratives are specific in that each unit, or event, occurs in a particular way. Similarly, narratives are fixed, rather than adaptable. Further distinguishing them from schemas, narratives can be encoded in an episode, rather than being composed of commonalities extracted from multiple episodes. (Ghosh Gilboa, p. 110)

In this definition, narratives embody the complexity schemas fail to encompass via particulars. As the previous chapter considered, the organization of particulars into meaningful, rememberable structures depends on a variety of factors, including temporal structures emerging from patterns of dynamic, relational valuation. Schema that facilitate an understanding of complex events and that foster effective predictive simulations may be organized around processing patterns rather than the analogy-> association->prediction circuit. That is, we process experience into chunked event structures via schema that create boundary definitions at crucial break points (such as, as mentioned above, equilibrium-disequilibrium-equilibrium patterns, or action-fact structures).

Narratives depend deeply upon schematic network elements. For example, in Elman and McRae's connectionist figuration of event processing network architecture, perception is attuned to 'Agents, Actions, Patients, Instruments, Contexts, and Recipients,' providing evidence of the selection and sequencing processing based on relational, dynamic valuation that figures into event definition (see Chap. 6). I speculate further that these general schematic categories have either situation-specific variants or embedded subschema that attune the subject to experiencing certain situations in terms of weak narrativity.

To summarize, current research on schema, while underdeveloped regarding the relation between schemas and narratives, suggests the following: narratives are schema-dependent, emergent in experiential processing, component to gist processing, and refined in memory processes.

Gists appear to be vital to another function which is vexing for schema theory: processing ill-defined problems via scenario simulation. In "ill-defined problems," there is "no specific script or solution path that can be applied" (Sheldon et al. 2011, p. 2). For ill-defined, or "open-ended" problems, because there's no solution to store semantically, "it [is] likely that solutions to such problems…rely on an event simulation strategy" (p. 2). Sheldon and colleagues cite findings (reviewed in Chap. 6) that episodic memory is used to imagine future events to hypothesize that episodic memory processes are implicated in solving ill-defined problems (*ibid*., 8), particularly because of the flexibility and "relational binding" capabilities of episodic memory. The mechanisms that initiate and perform these simulations are speculative but might include analogical processing at the beginning to identify similar problems from the past, the associated solutions to which would provide material from which solution simulations might be built (p. 8). Beatty, Morgan, and Wise emphasize that narrative simulations of 'counterfactual' events and structured around outcomes help reveal the contingency (and therefore meaning) of actual events. Additionally, problem-solving via simulation works by creating a phenomenology of feeling associated with particular scenarios, allowing us to 'pre-feel' possible experiences (Schacter et al. 2015).

Event simulation requires that subjects conceive of an outcome derived from beginning and ending value conditions of selected actors. Presumably, the repertoire of potential value conditions has certain primitives (including conditions of power in relation to the perceiving subject—see Chap. 7, Sect. 7.9) as well as culturally mediated conditions available in schema for these agents. The outcomes, too, may have primitives and schema ready for analogical processing. However, the processing of their dynamism and relationality (and the dynamics of their relations) requires a capacity for complex time-extended imagining: the ability to coordinate multiple agents and contextual factors into a meaningful sequence, resulting in an event (or chain of events). For a more detailed argument, (see Chap. 6).

The phenomenology associated with simulation (as a 'pre-feeling' of possible experiences) reflects how valuation involves a primitive concatenation of affective, embodied, and anticipatory value elements. Gists derive from high salience elements in the experiential field. Refining these elements into gists after experience must involve some selection and binding of these values via associative linkages to other experiences and schema. That is, gists are complexes of valuative elements—rather than simple unvalued particulars—with deep links to the outcome structures of narratives, and which are refined via (and which in turn contribute to) schemas or connectionist networks. More simply put, particulars from our experiences have emergent meanings that shape our general understanding while aligning our memories into flexible narrative patterns. Our reflective experiences of self and others necessarily bear the traces of gists in the schema we bring to bear on experience and the narrative processing we employ to recollect and anticipate it.

8.5 Self, Episodic Autobiographical Memory, and Social Cognition

Having considered how intermediary processing might involve narrative features, we will now turn to how weakly narrative cognition might interact with more strongly narrative aspects of selfhood, consciousness, and social cognition.

In "Narrative Modes of Consciousness and Selfhood" Keith Oatley synthesizes different approaches to narrative and selfhood. He proposes that people possess "narrative consciousness" in the form of a "unified narrative agent" (p. 378). He "explore[s] [following David Velleman] the idea of a conscious, unified self, based on functional properties of narrative" (p. 380). Oatley's views resonate with ideas traced earlier in this and other chapters. Specifically, narrative qualities, including action plans based upon goals and intentions, emotions as forms of self-feedback, and coherence based upon meaning are critical to the emergence of a central organizing agency, a self (pp. 383–4). These qualities, moreover, permit communication of our own mental states with others, enabling us to receive and process like communications, and to formulate theories of others' states of mind and self while unconsciously, and sometimes consciously, refining our own (pp. 385–6). Thus, we construct a self analogously to a novelist constructing a character, but "improvising as we go along" (p. 386). This self is "embodied … [and] accomplishes things in the world and interacts with others whom we assume are constituted in a way that is much like our self" (p. 386).

Oatley views selfhood in terms of a multiplicity of both 'low-level' processes and higher order personal and social processes. First, consciousness has four aspects: "simple awareness, the stream of inner consciousness, conscious thought as it may affect decisions and actions, [and] consciousness of self-with-other" (p. 377). These aspects roughly anticipate the sections that follow. Second, Oatley proposes multiple roles for emotions: as a "primary form of Helmholzian consciousness, … [as] frames or scripts for inter-individual relationships, … [as] caused by disruptions of action or expectancy" and as the basis, when combined with language, for "social cognition … based on narrative-like simulations" (p. 390). Third, narrative capacities are developmental. Finally, 'modern narratizing consciousness' is the product of historical changes in human interaction wrought by economic, political, and technological developments. Overall, Oatley's claims are grounded in the narrative quality of simulation.

In the previous section and the previous two chapters, we considered how more recent research into 'low-level' and intermediary processes might contribute to a sense of weak narrativity in experience. Recent work on episodic autobiographical memory (EAM) has shed some light on the possible relations between various notions of selfhood while illuminating mechanisms by which narrative structures form, circulate, and interact with semantic knowledge. The next section reviews some of this research before considering the development of narrative capacities in childhood and the role of narrative in folk psychology.

8.6 Selfhoods and Episodic Autobiographical Memory

How might recent research into memory inform theories about the narrativity of self? Markowitsch and Stanilou (2011) review the many different notions of self described in cognitive research: "narrative self," "self as containing conceptual knowledge of one's own personality traits," "self as… agent," "core self," "proto-self," and "feeling self" (p. 24). Amid this multiplicity, they seem to agree with Northoff's et al. (2006) assessment that it is unclear whether these are relevant only within specific cognitive functions (including memory) or whether there can be a unified theory of self. Several authors they cite (Feinberg and Keenan 2005; Panksepp 1998; Damasio 1999; Gallagher 2000) offer hierarchical notions of nested forms of self.

Autobiographical memory serves selfhood in its global forms—supporting a sense of personal identity within (and in relation to) social groups and the values and experiences that bind them. Autobiographical memory contributes to the time-extended, contextually complex idea of an ever-developing 'life story.'

The relation between episodic memory and autobiographical memory, then, is crucial to understanding how weak narrativity in general experience and reflective notions of narrative selfhood might be linked. Yet autobiographical memory is connected to semantic processes that also seem fundamental to selfhood. Martin Conway (2005) describe ‘autobiographical knowledge’ as distinctive from episodic memories (p. 589).

Autobiographical knowledge consists of context-free “conceptual, generic, and schematic knowledge about one’s life” (p. 589). By contrast, episodic memories are ‘experience-near’ “psychological or cognitive summaries of experience … derive[d] from mental models of ‘online’ experience configured in … the episodic buffer” (p. 589). According to this model, autobiographical knowledge manifests in a hierarchy capped by “overarching lifetime periods or themes” with an intermediate level of “repeated categories of events (i.e., every Thanksgiving) or temporally extended events (e.g., a picnic or vacation)” (Holland and Kensinger 2010, p. 4).

At the bottom of this hierarchy, autobiographical memories provide episodic referents for autobiographical knowledge. Among episodic memories, autobiographical memories are reinforced and retained because of their relevance to autobiographical knowledge and goals, as well as because of their emotional nature: “When we are cued to retrieve an autobiographical memory, we begin our search at the intermediate, general level and then move to retrieving more specific information” (p. 4). Autobiographical memory retrieval is *reconstructive*.

Thus, autobiographical functions are distributed along a three-tiered hierarchy of autobiographical knowledge, intermediate knowledge, and autobiographical memories. The structure of these processes provides some indication of the nature of our ‘life stories.’ Global and intermediate level “personal narratives” are by their nature dynamic, incorporating particularly important events—especially emotional events—through “enhanced rehearsal” (p. 19). Life narratives are continuously subject to reconstruction and supplementation, particularly in relation to “intermediate level” understandings, though sometimes via particularly impactful “flashbulb memories.”

This model is reminiscent of some of the issues we considered regarding the interaction of schema with perceptual data in event processing. (see Chap. 6). Yet here, the transition between recollected events and self-schema is mediated, and indeed conducted, by salient types of events, extended events, and particularly impactful individual events. In other words, memory reconstruction is initiated from knowledge of past experiences that, whether cumulatively or because of their extension or salience, have been preserved as more abstract summaries likely characterized by strong valuations. As discussed earlier (see Chaps. 6 and 7), the “whats” of events are refined in the course of relational weighting and temporal bounding, processes that draw upon valuations of various kinds (affordances, peak-end and equilibrium-disequilibrium-equilibrium patterns, identification and familiarity, etc.). Thus, this intermediate-level knowledge includes valuative elements. Valuation also characterizes autobiographical knowledge in which overarching themes and time periods reflect a sense of one's present relation to the past, and particularly to who one was in the past.

This is a key source of our commonsense notion of selfhood: rather than being a fixed substance, it is the ongoing self-evaluative processing by the subject of the self in present experiences in relation to past experiences, and vice versa.

This dynamism is partly due to the dynamism of a subject's immediate, intermediate-term, and long-term goals. Which goals have priority, particularly regarding attentional resources, provides a key index of self which the subject can introspect. These goals, as well as judgments about past and future likelihood of achieving them, are crucial, and not necessarily implicated in narrative processing. In addition, these self-generated self-valuations are (obviously) socially mediated. By considering one's own goals and achievement in relation to others', one calibrates one's own self-valuations.

Again, these forms of autobiographical knowledge about the self are not necessarily produced narratively. Yet, it is worthwhile to distinguish them from a commonsense understanding of traits. Whereas traits are propositional, goal-based valuations are temporal. As a result, the latter envisions selfhood in terms of ongoing comparison of the present to the past. Present experiences, in other words, contribute to self-evaluation in terms of relative valuations of current self-states versus self-states in past similar experiences. This modifies other metrics of self-valuation. To give a simple example, if one is a slow and messy knitter in relation to one's knitting group, this negative valuation may be attentuated by comparing one's current knitting achievements to past achievements. Of course, self-valuation also occurs in terms of present self-states in relation to projected future self-states. So one's present knitting accomplishment might, for example, be devalued if one is continuously fantasizing about future knitting accomplishments after attending a pricey knitting seminar. In both of these forms of temporal comparison inchoate narratives may support self-valuation (in the first example, a positive self-valuation based upon progress; in the second example, a complex self-valuation that diminishes self-value in the present but in anticipation of a future higher self-valuation).

Such time-extended self-valuation provides a useful feedback loop which enables the subject to modulate behavior, especially in social contexts. Yet comparative valuation also powerfully feeds into ideas about self. For example, if one makes a joke at a work event that is met with snubs, one may draw conclusions about the appropriate behavior expected at this and like functions and moderate one's comportment accordingly. Yet if one finds a co-worker then successfully making jokes, one's self-valuation, either globally or regarding one's status in this social milieu, may change (i.e., in this example, one reflects 'maybe the response is not a result of the context; rather, it's about my ability to be funny'). The point is partly that self-valuation shapes how we encounter experiences at (perhaps unobserved) set levels. These are revealed when experiences provide feedback that leads us to modulate them. In this way a nascent sense of self-consciousness emerges.

Time-extended, relationally-sensitive self-valuation is thus component to experiences of selfhood. Autobiographical knowledge of the features of the self can be flexible. Rather than offering a fixed set of traits around which autobiographical narratives are shaped, our selves are embedded in small life stories that can be contingent, iterative, and malleable.

This source of malleability requires some rethinking of approaches to narrative identity. When we consider models of the narrative self which posit narrative construction as an essential activity for generating meaning and coherence, we find, in contrast to much of the established research, that narrative construction is in part facilitated by the self-valuations emerging from the weakly narrative nature of experience. Most models do not recognize this feature, but rather attend to another important feature of narrativity—how high-level narratives or narrator functions shape immediate experiences.

Some theories, such as Rubin et al. (2003), hold that narrative both plays an important role in autobiographical memory and can be non-linguistic. For this view, narrative processing might play an important role in connecting various processing "levels" of consciousness. For example, consider the model of "flows across multiple levels of consciousness" Markowitsch and Stanilou attribute to Davies (1996), among others:

> multiple levels of consciousness and unconsciousness, in an ongoing state of interactive articulation as past experience infuses the present and present

> experience evokes state-dependent memories of formative interactive presentations. Not an onion, which must be carefully peeled, or an archaeological site to be meticulously unearthed and reconstructed in its original form, but a child's kaleidoscope in which each glance through the pinhole of a moment in time provides a unique view…: determined by way of infinite pathways of interconnectedness. (p. 197)

As the discussion of event segmentation (Chap. 6), episodic memory (Chap. 5, Sect. 5.2 and Chap. 6, Sect. 6.5), and the intersections of episodic and autobiographical memory have emphasized, however, the "moment in time" view of these images is reductive. Bounded temporal extensions, segmented into basic 'chunks' of experience, subject to revision and rehearsal, are emergent 'primitives' of our experiences for which Davis, Markowitsch, and Stanilou's "infinite pathways of interconnectedness" must account. The position I've argued is that, rather than temporal markers becoming simply semantic facts incorporated into schema (though they are likely processed in this manner), they are inextricably shaped by the cognitive-emotional and relational forms of valuation that allow for the priming and prediction of subsequent experiences.

Such a view requires that EAM, and particularly the ability to mentally time travel, not be dependent on language. According to Markowitsch and Stanilou, although "language and mental time travel might have co-evolved,… mental time traveling might have been rooted in a form of embodied cognition" (p. 26).

Narrative may have another relation to episodic memories as well. Milivojevic, Varadinov et al. (2016) hypothesize that

> narratives may provide a general context, unrestricted by space and time, which can be used to organize episodic memories into networks of related events. However, it is not clear how narrative contexts are represented in the brain. Here, we test the novel hypothesis that the formation of narrative-based contextual representation in humans relies on the same hippocampal mechanisms that enable formation of spatiotemporal contexts in rodents. (p. 12412)

So it is possible that the "higher order" organization of episodic memories draws on the same processing mechanisms involved in event segmentation.

As we continue to revise our understanding of how the hippocampal region facilitates future simulation, predictive processing, and affective cognition, we may clarify how narrative processing occurs, and perhaps develop a novel taxonomy of types of interrelated forms of narrative processing distinguished by the role intention, iteration and rehearsal, language, and experience nearness and distance play in forms of cognition we have described.

The model of (higher order) narrative I am proposing, then, is an iterative processing logic that draws upon and informs weak narrativity in experience. Strong narrativity may emerge in various forms in relation to this processing and may involve processes of narrative rehearsal that shape particular accounts of episodes and larger 'stories' of experience mutually.

Theories of memory, self, and narrative development in childhood provide some support for this idea. In the model of the emergence of autobiographical memory offered by Nelson and Fivush, "narrative structure and content" develops around ages 3–5, after the establishment of a "core self" and in feedback with the emergence of increasingly autonomous temporal concepts, complex language concepts, self-representation, and theory of mind. It co-develops roughly with the establishment of episodic memory and overlaps with the establishment of autobiographical memory.[1]

So, if the ability to form episodic memories is related to the emergence of a theoretical weak narrativity in experience, then these malleable narrative forms help drive autobiographical self-understanding.

One key area of interest for researchers that involves "higher order" cognition is how narratives operate in social cognition. This chapter concludes by considering how narrative processing might be crucial to an essential way we navigate the social world: through "theory of mind."

8.7 Social Cognition and Narrative

Hasson, Furman et al. (2008) note a deep correlation between brain regions involved in social cognition and episodic memory formation for narratives.[2] Viewing narrative as a practice for making inferences about targeted social agents' beliefs, motivations and values is part of a debate over the workings of folk psychology. In a review of this debate, Hutto and McGivern (who advocate a narrative processing account of theory of mind) contrast "theory theory" with their narrative practice hypothesis (NPH). "Theory theory" involves having a general set of ideas about mind and its exterior signs providing the basis for our folk psychology, whereas the narrative practice hypothesis postulates that the consumption and creation of narratives provide this basis. Folk psychology, in the NPH, involves a simulation of experience from the perspective of another in a narrative logic, which enables *ad hoc* particulars, situations, and contexts to be combined into meaningful form. For Hutto and McGivern, "theory theory" struggles to explain how its general principles are practically realized in particulars.

Standard "theory theory" is propositional. However, Hutto and McGivern review a version promoted by Maibom (2009) as "model theory," in which the centrality of practice (that is, realization in particulars) is recognized (Hutto and McGivern, rpt. from Maibom (p. 361)). "Model theory" evokes definitional concerns about narrative as a "thing" or a form of practice: "this massively softer way of construing theory theory," Maibom (2009) argues, "accounts *not just as well as, but better than*, narrativity theory for the fact that our folk psychological explanations appear to contain, or form part of, narratives" (Maibom 2009, p. 361, emphasis added). Model theory, then, points to how narrative forms can emerge from diverse forms of practice which themselves deploy models that render a narrative character in experience. That is, instead of narrativity deriving entirely from consumed, artifactual narratives or consciously created narratives, narrativity inheres in or emerges from the sensemaking process.

This view fits well with Dewhurst's (2017) speculation that if we adopt a predictive processing framework we'll need to develop "a hybrid theory that includes elements of both theory-theory and simulation theory," which might include "narratives and social norms," including folk psychological narratives operating "as a form of cognitive scaffolding" (p. 9). Thus, Dewhurst does not see folk psychology as impossible to "reconcile with predictive processing," though he believes we need a "novel conceptual taxonomy that more accurately reflects the structure of cognition and allows us to move beyond the limitations of folk psychological discourse" (p. 11).

In the previous two chapters, some perceptions driven by affective, relational, and dynamic valuations with emergent temporal boundaries are characterized by "weak narrativity." This model constitutes one attempt to add to our conceptual repertoire.[3] If the simulation processes characteristic of predictive processing depend upon weak narrativity (see Chap. 6), then perhaps the capacity to simulate another's mindset has related origins (see Hutto and McGivern; also Stueber 2012).

The limitation of this view, according to Hutto and McGivern (citing Medina 2013; Moran 1994), is that simulation theory lacks the 'hot' character of the phenomenology of happening characteristic of theory of mind.

For Hutto and McGivern, traditional accounts of "narrative understanding" and "simulation" lack the feeling of being viscerally moved—that is, the feeling of active engagement. Citing Medina, they argue that "what drives our hot narrative engagements is neither self-projective simulation, nor simulating or pretending to believe a set of propositions; rather, 'the central difference between emotionally engaged and emotionally disengaged imagination has to do with the mode of imagining, not with its content'". Hutto and McGivern's point resonates with research by Hassabis and Maguire that scene construction, rather than self-projection, links episodic memory with "other cognitive functions such as episodic future thinking, navigation, and theory of mind" (2016, p. 299).

Drawing upon conventional elements of literary narrative, Medina believes "mode of imaging" determines the "hot" character of theory of mind. I'm not convinced this is sustainable—instead, I believe these resources ("figuration, allusion, rhythm, repetition, assonance, and dissonance") in the arts create a 'hot' character in artistic artifacts (including the visual arts and music) that simulates the (generally) more weakly hot character of our experiences. Rather, as Hutto and McGivern seem to suggest, this hot character is in large part wrought by the enactive and other valuative elements of both real-time and imagined experiences.

One way of explaining this is to consider what sensitivities are developed through long experience with processing narratives: "by developing our narrative understanding through [consuming narratives] … we become sensitive to a variety of possible perspectives that may be adopted on events, including—especially—cognitive, emotional, and evaluative perspectives that may diverge from our own" (n.p.).

Of course, our apprehension of another's "perspective" via narrative involves a complex synthesis of overt characterization (whether from a narrator's, the character's, or another's perspective) and inferences about the principles, sense of self or self-narrative, 'hot' nature (or phenomenology), and priors that inform particular reactions to particular events. In other words, even in the consumption of artifactual narratives, perspective involves a great deal of inferential reasoning about the interior dynamics of another that result in particular and—if the inference is successful—relatively predictable valuations of experience. In this, as in the discussions of episodic memory and schema, as well as the discussion of self, this involves modeling that helps predict patterns of valuation.

In Theory of Mind, then, we seek to understand the valuative predilections that comprise the givenness of experience for another. As many theorists have noted, TOM provides a ready means of social learning that helps us contextualize, evaluate, and revalue our own values. It is crucial to self-reflection.

In engaging in this inferential process in non-narrative contexts, particularly banal social experiences, we may in part experience it as a set of propositions that make up our understanding of another. For example, note the valuative aspects of the following: "Tom does not like cats, and he is a fan of avant-garde jazz and expensive wine." We might develop more abstract ideas about the other's character from these facts—Tom is a 'highbrow' and maybe a snob—from which we might create, associatively, speculative propositions ("Tom probably only dates highly educated people"). We may also develop a set of causally related insights: "Tom has allergies, and maybe this explains why he doesn't like cats." Yet as a model of Tom, this is sterile, "cool" stuff.

However, if we evaluate Tom narratively (instead of propositionally)—"he flirted with but ultimately snubbed my friend Kelly" or "he confessed that he grew up impoverished in a house with alcoholic parents and more than 20 cats," then our engagement may become warmer—we may have visions of Tom's life that connect to our understanding of Tom in the present day.

Tom's supposed propositional characteristics are now placed into relational, real-time valuative contexts. And in this form, we may be more likely to simulate Tom's experiences or an encounter with him. In other words, processing what we learn about Tom's experiences and actions in narrative terms engages our emotional faculties and leads to a greater sense of engaging his perspective. This cues our future responses to him, though our understanding is subject to alteration via new knowledge, experiences, or exposure to other people's perspectives on him. Theory of mind is greatly enhanced by 'hot' weakly narrative simulation.

Theory of mind reflects social processes that may happen automatically, but have an important link to highly reflective forms of narrative that provide a form of explanation that uses contingent particulars to explain the nature of events and the actors that produce them. We reviewed how the logic of narrative affords explanation of heterogenous elements in a variety of disciplines in Chap. 2. As John Beatty memorably puts the matter, "Narratives tell us, and sometimes in a manner that not only represents the world but makes sense of it, by leading us from the contingent past to outcomes that are contingent upon their past" (p. 41).

This chapter has considered multiple ways narrative might play a role in a variety of forms of "higher order" cognitive processing. Viewing narrative from a processing perspective, based upon dynamic valuations, allows us to speculate about a more or less vital role for narrative in our experiences, our identities, and our social intuitions.

Notes

1. Note, though, that complex language representation and narrative structure and content are reciprocally influential. The theory I'm advancing is not that language does not support narrative representation. However, it remains necessary to account for how complex language facility shapes capacities for narrative structure and content.
2. They note that "given that much of what we encounter and value on a daily basis is fundamentally social, these findings highlight the important role that social cognitive processes play in episodic memory formation" (p. 458).
3. There are others. For example, Lisa Quadt's notion of Embodied Social Inference sees embodied cognition as crucial to social cognition.

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